

Praktikum 1 - Struktur Sistem Operasi



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Praktik 1.1: Menjelajahi Layanan OS

Praktik 1.2: Program C - Menggunakan Layanan OS

```
opank@MSI: ~  
Mmio stale data:      Not affected  
Reg file data sampling: Vulnerable: No microcode  
Retbleed:             Mitigation; Enhanced IBRS  
Spec rstack overflow: Not affected  
Spec store bypass:    Mitigation; Speculative Store Bypass disabled via  
                      prefill  
Spectre v1:           Mitigation; usercopy/swaps barriers and __user po  
                      inter sanitization  
Spectre v2:           Mitigation; Enhanced / Automatic IBRS; IBPB condit  
                      ional; RSB filling; PBRSE-eIBRS SW sequence; BHI B  
                      HI_DIS_S  
Srbds:                Not affected  
Tsx async abort:      Not affected  
opank@MSI:~$ lsblk  
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS  
sda         8:0    0 388.4M  1 disk  
sdb         8:16   0 186M    1 disk  
sdc         8:32   0    1G    0 disk [SWAP]  
sdd         8:48   0    1T    0 disk /mnt/wslg/distro  
/  
opank@MSI:~$ sudo os_services.c  
[sudo] password for opank:  
opank@MSI:~$ sudo os_services.c: command not found  
opank@MSI:~$ 123  
123: command not found  
opank@MSI:~$ sudo os_services.c  
sudo: os_services.c: command not found  
opank@MSI:~$ sudo apt install gcc  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu gcc-13  
  gcc-13-base gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu libaom3  
  libasan0 libatomic1 libc-dev-bin libc-devtools libc6-dev libcc1-0  
  libcrypt-dev libde265-0 libgcc-13-dev libgd3 libgomp1  
  libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265  
  libheif1 libhwasan0 libisl23 libitm1 liblsan0 libmpc3 libquadmath0  
  libtsan2 libubsan1 libxpm4 linux-libc-dev manpages-dev rpcsvc-proto  
Suggested packages:  
  cpp-doc gcc-13-locales cpp-13-doc gcc-multilib make autoconf automake  
  libtool flex bison gdb gcc-doc gcc-13-multilib gcc-13-doc  
  gdb-x86-64-linux-gnu glibc-doc libgd-tools libheif-plugin-x265  
  libheif-plugin-fmpgdec libheif-plugin-jpegdec libheif-plugin-jpegenc  
  libheif-plugin-j2kdec libheif-plugin-j2kenc libheif-plugin-rav1e  
  libheif-plugin-svtenc  
The following NEW packages will be installed:  
  cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu gcc gcc-13  
  gcc-13-base gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu libaom3  
opank@MSI: ~  
passwd: password updated successfully  
Installation successful!  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
opank@MSI:~$ uname -a  
Linux MSI 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025  
opank@MSI:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 24.04.1 LTS  
Release:        24.04  
Codename:       noble  
opank@MSI:~$ ps aux | head -20  
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND  
root         1  0.0  0.3 21708 12464 ?        Ss   22:37   0:00 /sbin/init  
root         2  0.0  0.0   3072  1792 ?        Sl   22:37   0:00 /init  
root         7  0.0  0.0   3088  1920 ?        Sl   22:37   0:00 plan9 --control-socket 7 --l  
root        50  0.0  0.4 50440 15616 ?        S-s  22:37   0:00 /usr/lib/systemd/systemd-jou  
root       103  0.1  0.1 25260  6000 ?        Ss   22:37   0:01 /usr/lib/systemd/systemd-ude  
systemd+   109  0.0  0.3 21452 12672 ?        Ss   22:37   0:00 /usr/lib/systemd/systemd-res  
systemd+   110  0.0  0.1  91020  7680 ?        Ssl  22:37   0:00 /usr/lib/systemd/systemd-tin  
root       160  0.0  0.0   4236  2560 ?        Ss   22:37   0:00 /usr/sbin/cron -f -P  
message+   161  0.0  0.1   9620  4992 ?        Ss   22:37   0:00 @dbus-daemon --system --addr  
root       168  0.0  0.2  17960  8448 ?        Ss   22:37   0:00 /usr/lib/systemd/systemd-log  
root       171  0.0  0.3 1756096 12160 ?       Ssl  22:37   0:00 /usr/libexec/wsl-pro-service  
root       186  0.0  0.0   3160  1920 hvcs0 Ss+  22:37   0:00 /sbin/agetty -o -p -- \u --r  
syslog     190  0.0  0.1 222508  5248 ?        Ssl  22:37   0:00 /usr/sbin/rsyslogd -n -iNONE  
root       197  0.0  0.0   3116  1792 tty1    Ss+  22:37   0:00 /sbin/agetty -o -p -- \u --r  
root       198  0.0  0.5 107012 22780 ?        Ssl  22:37   0:00 /usr/bin/python3 /usr/share/  
root       267  0.0  0.0   3076   896 ?        Ss   22:37   0:00 /init  
root       268  0.0  0.0   3092  1024 ?        S   22:37   0:00 /init  
opank     269  0.0  0.1   6072  5120 pts/0    Ss   22:37   0:00 -bash  
root      270  0.0  0.1   6820  4480 pts/1    Ss   22:37   0:00 /bin/login -f  
opank@MSI:~$ free -h  
              total        used        free      shared  buff/cache   available  
Mem:          3.7Gi          419Mi        3.2Gi        3.5Mi        149Mi        3.3Gi  
Swap:          1.0Gi           0B          1.0Gi  
opank@MSI:~$ lscpu  
Architecture:        x86_64  
CPU op-mode(s):      32-bit, 64-bit  
Address sizes:        39 bits physical, 48 bits virtual  
Byte Order:           Little Endian  
CPU(s):               16  
On-line CPU(s) list: 0-15  
Vendor ID:            GenuineIntel  
Model name:           12th Gen Intel(R) Core(TM) i7-12650H  
CPU family:           6  
Model:               154  
Thread(s) per core:   2
```

2. User and Operating-System Interface

Praktik 2.1: Command Line Interface (CLI)

Praktik 2.2: Program C - Shell Sederhana Buat file simple_shell.c:

```
opank@MSI: ~$ pwd
/home/opank
opank@MSI: ~$ ls
os_services  os_services.c  test_file.txt
opank@MSI: ~$ cd /tmp
opank@MSI: /tmp$ mkdir test_dir
opank@MSI: /tmp$ cd test_dir
opank@MSI: /tmp/test_dir$ cd opank@MSI
-bash: cd: opank@MSI: No such file or directory
opank@MSI: /tmp/test_dir$ ls
total 0
opank@MSI: /tmp/test_dir$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.3 21532 12544 ?        Ss   22:36   0:01 /usr/lib/
root         2  0.0  0.0   3972  1792 ?        Sl   22:36   0:00 /init
root         7  0.0  0.0   3888  1920 ?        Sl   22:36   0:00 plan9 --c
root        50  0.0  0.4 50440 15744 ?        S<s  22:36   0:00 /usr/lib/
root       103  0.1  0.1 25264  6400 ?        Ss   22:36   0:03 /usr/lib/
systemd+  109  0.0  0.3 21452 12672 ?        Ss   22:36   0:00 /usr/lib/
systemd+  110  0.0  0.1 91020  7680 ?        Ssl  22:36   0:00 /usr/lib/
root       160  0.0  0.0   4236  2560 ?        Ss   22:36   0:00 /usr/sbin
message+  161  0.0  0.1  9520  5120 ?        Ss   22:36   0:00 @dbus-dae
root       168  0.0  0.2 17960  8448 ?        Ss   22:36   0:00 /usr/lib/
root       171  0.0  0.3 1756096 12288 ?       Ssl  22:36   0:00 /usr/libe
root       186  0.0  0.0   3160  1920 hvc0     Ss+  22:36   0:00 /sbin/age
systlog   190  0.0  0.1 222508  5248 ?        Ssl  22:36   0:00 /usr/sbin
root      197  0.0  0.0   3116  1792 tty1     Ss+  22:36   0:00 /sbin/age
root      198  0.0  0.5 107012 22780 ?        Ssl  22:36   0:00 /usr/bin/
root      267  0.0  0.0   3076   896 ?        Ss   22:36   0:00 /init
root      268  0.0  0.0   3092  1024 ?        S   22:36   0:00 /init
opank     269  0.0  0.1  6072  5120 pts/0    Ss   22:36   0:00 -bash
root      270  0.0  0.1  6020  4400 pts/1    Ss   22:36   0:00 /bin/login
opank     302  0.0  0.2 20308 11392 ?        Ss   22:36   0:00 /usr/lib/
opank     303  0.0  0.0  21148  3520 ?        S   22:36   0:00 (sd-pam)
opank     318  0.0  0.1  6056  5248 pts/1    S+   22:36   0:00 -bash
polkitd   855  0.0  0.1 308160  7680 ?        Ssl  22:58   0:00 /usr/lib/
opank    2274 100  0.1  8200  4224 pts/0    R+   23:23   0:00 ps aux
opank@MSI: /tmp/test_dir$ cd test_dir
-bash: cd: test_dir: No such file or directory
opank@MSI: /tmp/test_dir$ ls
opank@MSI: /tmp/test_dir$ ls -l | grep "d"
d
opank@MSI: /tmp/test_dir$ echo "Hello World" > output.txt
opank@MSI: /tmp/test_dir$ cat output.txt
Hello World
opank@MSI: /tmp/test_dir$ cat output.txt >> output.txt

opank@MSI: ~$ nano simple_shell.c
opank@MSI: ~$ ls
os_services  os_services.c  simple_shell.c  test_file.txt
opank@MSI: ~$ gcc -o simple_shell simple_shell.c
opank@MSI: ~$ ./simple_shell
myshell>
myshell> ls
os_services  os_services.c  simple_shell  simple_shell.c  test_file.txt
myshell> cd -
execvp failed: No such file or directory
myshell> ^X^C
opank@MSI: ~$ nano process_syscalls.c
opank@MSI: ~$ ls
os_services  process_syscalls.c  simple_shell.c  test_file.txt
opank@MSI: ~$ nano info_syscalls.c
opank@MSI: ~$ nano file_syscalls.c
opank@MSI: ~$ ls
file_syscalls.c  os_services.c  process_syscalls.c  simple_shell.c  test_file.txt
opank@MSI: ~$ gcc -o process_syscalls process_syscalls.c
opank@MSI: ~$ gcc -o file_syscalls file_syscalls.c
opank@MSI: ~$ gcc -o info_syscalls info_syscalls.c
opank@MSI: ~$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 2478
Parent waiting for child (PID: 2479)...
Child PID: 2479
Child's Parent PID: 2478
Child executing 'ls -l' command...
total 112
-rwxr-xr-x 1 opank opank 16352 Oct 15 23:43 file_syscalls
-rw-r--r-- 1 opank opank 1481 Oct 15 23:40 file_syscalls.c
-rwxr-xr-x 1 opank opank 16304 Oct 15 23:43 info_syscalls
-rw-r--r-- 1 opank opank 810 Oct 15 23:42 info_syscalls.c
-rwxr-xr-x 1 opank opank 16488 Oct 15 23:07 os_services
-rw-r--r-- 1 opank opank 1650 Oct 15 23:05 os_services.c
-rwxr-xr-x 1 opank opank 16360 Oct 15 23:43 process_syscalls
-rw-r--r-- 1 opank opank 1103 Oct 15 23:36 process_syscalls.c
-rwxr-xr-x 1 opank opank 16528 Oct 15 23:33 simple_shell
-rw-r--r-- 1 opank opank 1347 Oct 15 23:32 simple_shell.c
-rw-r--r-- 1 opank opank 23 Oct 15 23:07 test_file.txt
Child exited with status: 0
opank@MSI: ~$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
```

Praktik 3.2: File Management System Calls

```
opank@MSI: ~  
opank@MSI:~$ nano file_syscalls.c  
opank@MSI:~$ ls  
file_syscalls.c  os_services.c      simple_shell    test_file.txt  
os_services     process_syscalls.c  simple_shell.c  
opank@MSI:~$ nano info_syscalls.c  
opank@MSI:~$ ls  
file_syscalls.c  os_services  process_syscalls.c  simple_shell.c  
info_syscalls.c  os_services.c  simple_shell        test_file.txt  
opank@MSI:~$ gcc -o process_syscalls process_syscalls.c  
opank@MSI:~$ gcc -o file_syscalls file_syscalls.c  
opank@MSI:~$ gcc -o info_syscalls info_syscalls.c  
opank@MSI:~$ ./process_syscalls  
=== Process Control System Calls ===  
Parent PID: 2478  
Parent waiting for child (PID: 2479)...  
Child PID: 2479  
Child's Parent PID: 2478  
Child executing 'ls -l' command...  
total 112  
-rwxr-xr-x 1 opank opank 16352 Oct 15 23:43 file_syscalls  
-rw-r--r-- 1 opank opank 1431 Oct 15 23:40 file_syscalls.c  
-rwxr-xr-x 1 opank opank 16384 Oct 15 23:43 info_syscalls  
-rw-r--r-- 1 opank opank 810 Oct 15 23:42 info_syscalls.c  
-rwxr-xr-x 1 opank opank 16488 Oct 15 23:07 os_services  
-rw-r--r-- 1 opank opank 1650 Oct 15 23:05 os_services.c  
-rwxr-xr-x 1 opank opank 16360 Oct 15 23:43 process_syscalls  
-rw-r--r-- 1 opank opank 1103 Oct 15 23:36 process_syscalls.c  
-rwxr-xr-x 1 opank opank 16528 Oct 15 23:33 simple_shell  
-rw-r--r-- 1 opank opank 1347 Oct 15 23:32 simple_shell.c  
-rw-r--r-- 1 opank opank 23 Oct 15 23:07 test_file.txt  
Child exited with status: 0  
opank@MSI:~$ ./file_syscalls  
=== File Management System Calls ===  
File descriptor: 3  
Bytes written: 25  
File closed  
Read from file: Hello from system calls!  
opank@MSI:~$ ./info_syscalls  
=== Information Maintenance System Calls ===  
  
Process ID: 2485  
Parent Process ID: 269  
User ID: 1000  
Group ID: 1000  
Current time: Wed Oct 15 23:45:11 2025  
Clock ticks since boot: 1718373315  
Sleeping for 2 seconds...  
Awake!  
opank@MSI:~$ systemctl list-units --type=service
```


5. Linkers and Loaders

Praktik 5.1: Proses Linking

Praktik 5.2: Static vs Dynamic Linking

Praktik 5.3: Membuat dan Menggunakan Shared Library

```
opank@MSI:~$ nano math_ops.h
opank@MSI:~$ ls
file_syscalls  math_ops.h  os_services.c  simple_shell.c
file_syscalls.c  my_daemon  process_syscalls  sudo
info_syscalls  my_daemon.c  process_syscalls.c  syscall_test.txt
info_syscalls.c  os_services  simple_shell  test_file.txt
opank@MSI:~$ nano math_ops.c
opank@MSI:~$ ls
file_syscalls  math_ops.c  os_services  simple_shell  test_file.txt
file_syscalls.c  math_ops.h  os_services.c  simple_shell.c
info_syscalls  my_daemon  process_syscalls  sudo
info_syscalls.c  my_daemon.c  process_syscalls.c  syscall_test.txt
opank@MSI:~$ nano main.c
opank@MSI:~$ ls
file_syscalls  math_ops.c  os_services  simple_shell  test_file.txt
file_syscalls.c  math_ops.h  os_services.c  simple_shell.c
info_syscalls  my_daemon  process_syscalls  sudo
info_syscalls.c  my_daemon.c  process_syscalls.c  syscall_test.txt
main.c  os_services  simple_shell.c
opank@MSI:~$ gcc -E main.c -o main.i
opank@MSI:~$ gcc -E math_ops.c -o math_ops.i
cc1: fatal error: math_ops.c: No such file or directory
compilation terminated.
opank@MSI:~$ nano math_ops.c
opank@MSI:~$ nano math_ops.c
opank@MSI:~$ gcc -E math_ops.c -o math_ops.i
opank@MSI:~$ gcc -S main.c -o main.s
opank@MSI:~$ gcc -S math_ops.c -o math_ops.s
opank@MSI:~$ gcc -c main.c -o main.o
opank@MSI:~$ gcc -c math_ops.c -o math_ops.o
opank@MSI:~$ gcc -c math_ops.o -o program
opank@MSI:~$ nm main.o
0000000000000000 T main
0000000000000000 U multiply
0000000000000000 U printf
opank@MSI:~$ nm math_ops.o
0000000000000000 T add
0000000000000018 T multiply
opank@MSI:~$ nm program
0000000000003dc8 d _DYNAMIC
0000000000003f08 d _GLOBAL_OFFSET_TABLE_
0000000000002000 b _IO_stdin_used
0000000000000000 w _ITM_deregisterTMCloneTable
opank@MSI:~$ ldd program
linux-vdso.so.1 (0x00007ffe9d787000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007fc5b1e00000)
/lib64/ld-linux-x86-64.so.2 (0x00007fc5b209d000)
opank@MSI:~$ gcc main.c math_ops.c -o program_static -static
opank@MSI:~$ gcc main.c math_ops.c -o program_dynamic
opank@MSI:~$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 opank opank 16K Oct 16 00:35 program_dynamic
-rwxr-xr-x 1 opank opank 767K Oct 16 00:35 program_static
opank@MSI:~$ ldd program_static
not a dynamic executable
opank@MSI:~$ ldd program_dynamic
linux-vdso.so.1 (0x00007fffdad1b000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007a49f9600000)
/lib64/ld-linux-x86-64.so.2 (0x00007a49f98bf000)
opank@MSI:~$ gcc -fPIC math_ops.c -o math_ops.o
opank@MSI:~$ gcc -shared -o libmathops.so math_ops.o
opank@MSI:~$ gcc main.c -L. -lmathops -o program_shared
opank@MSI:~$ LD_LIBRARY_PATH=. ./program_shared
5 + 3 = 8
5 * 3 = 15
opank@MSI:~$ readelf -d program_shared

Dynamic section at offset 0x2da8 contains 28 entries:
Tag              Type              Name/Value
0x0000000000000001 (NEEDED)           Shared library: [libmathops.so]
0x0000000000000001 (NEEDED)           Shared library: [libc.so.6]
0x000000000000000c (INIT)            0x1000
0x000000000000000d (FINI)            0x11e4
0x0000000000000019 (INIT_ARRAY)          0x2d98
0x000000000000001b (INIT_ARRAYSZ)        8 (bytes)
0x000000000000001a (FINI_ARRAY)          0x3da0
0x000000000000001c (FINI_ARRAYSZ)        8 (bytes)
0x0000000000000045 (GNU_HASH)           0x3db
0x0000000000000005 (STRTAB)              0x4b0
0x0000000000000006 (SYMTAB)              0x3d8
0x000000000000000a (STRSZ)               170 (bytes)
0x000000000000000b (SYMINFO)             24 (bytes)
0x0000000000000015 (DEBUG)               0x0
0x0000000000000003 (PLTGOT)              0x3fa8
0x0000000000000002 (PLTREL32)            72 (bytes)
0x0000000000000014 (PLTREL)              RELA
0x0000000000000017 (JMPREL)              0x660
0x0000000000000007 (RELA)                0x5a0
0x0000000000000000 (RELA32)             192 (bytes)
```

Praktik 6.2: Binary Format Analysis

```

[27] comment 0000000000000000 00000000 00000000
[28] systrab 0000000000000000 00000000 00000000
[29] strtab 0000000000000000 00000000 00000000
[30] shstrtab 0000000000000000 00000000 00000000
0000000000000000 0 0 0 1
0000000000000000 0 0 0 1
May be flags:
W (write), A (alloc), X (execute), M (merge), S (strings), I (info),
O (OS specific), C (compressed), u (unknown), o (OS specific), E (exclude),
D (debug), l (large), p (processor specific)
[31] .text 0x0000000000000000 0x0000000000000000 0x0000000000000000
ELF file type is DYN (Position-Independent Executable file)
Entry point 0x1000
There are 13 program headers, starting at offset 64

Program Headers:
Type           Offset             VirtAddr           PhysAddr
               #Bytes          Memsz              Flags              Align
PHDR           0x0000000000000000 0x0000000000000000 0x0000000000000000
INTERP         0x0000000000000018 0x0000000000000018 0x0000000000000018
               0x000000000000001c 0x000000000000001c R 0x1
[Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]
LOAD           0x0000000000000000 0x0000000000000000 0x0000000000000000
               0x0000000000000028 0x0000000000000028 R 0x1000
LOAD           0x0000000000000018 0x0000000000000018 0x0000000000000018
               0x0000000000000020 0x0000000000000020 R 0x1000
LOAD           0x000000000000001c 0x000000000000001c 0x000000000000001c
               0x0000000000000020 0x0000000000000020 R 0x1000
LOAD           0x0000000000000028 0x0000000000000028 0x0000000000000028
               0x0000000000000020 0x0000000000000020 R 0x1000
LOAD           0x0000000000000020 0x0000000000000020 0x0000000000000020
               0x0000000000000020 0x0000000000000020 R 0x1000
LOAD           0x0000000000000028 0x0000000000000028 0x0000000000000028
               0x0000000000000020 0x0000000000000020 R 0x1000
LOAD           0x000000000000002c 0x000000000000002c 0x000000000000002c
               0x0000000000000020 0x0000000000000020 R 0x1000
DYNAMIC        0x0000000000000018 0x0000000000000018 0x0000000000000018
               0x000000000000001c 0x000000000000001c R 0x8
NOTE          0x0000000000000018 0x0000000000000018 0x0000000000000018
               0x000000000000001c 0x000000000000001c R 0x8
NOTE          0x0000000000000018 0x0000000000000018 0x0000000000000018
               0x000000000000001c 0x000000000000001c R 0x8
GNU_PROPERTY  0x0000000000000018 0x0000000000000018 0x0000000000000018
               0x000000000000001c 0x000000000000001c R 0x4
GNU_EX_FRAME 0x000000000000001c 0x000000000000001c 0x000000000000001c
               0x000000000000001c 0x000000000000001c R 0x4

```

7. OS Design and Implementation 12 / 24 2025-10-06

Praktik 7.1: Kernel Module Sederhana

Praktik 7.2: Monitoring Kernel

```
opank@MSI:~$ nano hello_module.c
opank@MSI:~$ ls
file_syscalls.c  main.i  math_ops.o  process_syscalls.c  simple_shell.c
file_syscalls.c  main.o  math_ops.s  process_syscalls.c  sudo
hello_module.c  main.s  my_daemon.c  program              syscall_test.txt
info_syscalls.c  math_ops.c  my_daemon.c  program_dynamic      test_file.txt
info_syscalls.c  math_ops.c  os_services.c  program_shared
libmathops.so    math_ops.h  os_services.c  program_static
main.c           math_ops.i  os_specific.c  simple_shell
opank@MSI:~$ nano Makefile
opank@MSI:~$ ls
Makefile  main.c  math_ops.i  os_specific.c  simple_shell
file_syscalls.c  main.i  math_ops.o  process_syscalls.c  simple_shell.c
file_syscalls.c  main.o  math_ops.s  process_syscalls.c  sudo
hello_module.c  main.s  my_daemon.c  program          syscall_test.txt
info_syscalls.c  math_ops.c  my_daemon.c  program_dynamic  test_file.txt
info_syscalls.c  math_ops.c  os_services.c  program_shared
libmathops.so    math_ops.h  os_services.c  program_static
opank@MSI:~$ make
Command 'make' not found, but can be installed with:
sudo apt install make # version 4.3-4.1build1, or
sudo apt install make-guile # version 4.3-4.1build1
opank@MSI:~$ sudo insmod hello_module.ko
[sudo] password for opank:
insmod: ERROR: could not load module hello_module.ko: No such file or directory
opank@MSI:~$ sudo insmod hello_module.ko
insmod: ERROR: could not load module hello_module.ko: No such file or directory
opank@MSI:~$ make
Command 'make' not found, but can be installed with:
sudo apt install make # version 4.3-4.1build1, or
sudo apt install make-guile # version 4.3-4.1build1
opank@MSI:~$ sudo insmod hello_module.ko
insmod: ERROR: could not load module hello_module.ko: No such file or directory
opank@MSI:~$ sudo insmod hello_module.ko
insmod: ERROR: could not load module hello_module.ko: No such file or directory
opank@MSI:~$ make
Command 'make' not found, but can be installed with:
sudo apt install make # version 4.3-4.1build1, or
sudo apt install make-guile # version 4.3-4.1build1
opank@MSI:~$ sudo insmod hello_module.ko
insmod: ERROR: could not load module hello_module.ko: No such file or directory
opank@MSI:~$ lsmod | grep hello
opank@MSI:~$ dmesg | tail
[ 600.654257] mini_init (244): drop_caches: 1
[ 1369.799182] mini_init (244): drop_caches: 1
```


8. Operating-System Structure

Praktik 8.1: Eksplorasi Struktur Layering

Praktik 8.2: Microkernel vs Monolithic

Praktik 8.3: Program - Virtual File System

```
opank@MSI: ~  
Unpacking strace (6.8-0ubuntu2) ...  
Setting up strace (6.8-0ubuntu2) ...  
Processing triggers for nan-db (2.12.0-4build2) ...  
opank@MSI:~$ ps aux  
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND  
root         1  0.0  0.3 21956 13856 ?        Ss   Oct15   0:06 /usr/lib/systemd/systemd  
root         2  0.0  0.0   3872  1792 ?        Ss   Oct15   0:00 /init  
root         7  0.0  0.0   3888  1920 ?        Ss   Oct15   0:00 /bin/ls --control=socket 7  
root        50  0.0  0.4 66832 18844 ?        Ss+  Oct15   0:03 /usr/lib/systemd/systemd  
root       103  0.0  0.1 25264  6400 ?        Ss   Oct15   0:08 /usr/lib/systemd/systemd  
systemd+   189  0.0  0.3 21452 12800 ?        Ss   Oct15   0:00 /usr/lib/systemd/systemd  
systemd+   110  0.0  0.1 91020  7680 ?        Ssl  Oct15   0:00 /usr/lib/systemd/systemd  
root       160  0.0  0.0  4236  2560 ?        Ss   Oct15   0:00 /usr/sbin/cron -f -p  
message+   161  0.0  0.1  9696  5248 ?        Ss   Oct15   0:01 /usr/sbin/cron --system --  
root       168  0.0  0.2 13188  8448 ?        Ss   Oct15   0:00 /usr/lib/systemd/systemd  
root       171  0.0  0.3 175696 12880 ?       Ssl  Oct15   0:01 /usr/libexec/ssl-pro-sar  
root       186  0.0  0.0   3160  1920 hvco    Ss+  Oct15   0:00 /sbin/agetty -o -p -- \u  
syslog     190  0.0  0.1 222508  5248 ?       Ssl  Oct15   0:00 /usr/sbin/rsyslogd -n -i  
root       197  0.0  0.0   3116  1792 ttyl    Ss+  Oct15   0:00 /sbin/agetty -o -p -- \u  
root       198  0.0  0.5 197012 22780 ?       Ssl  Oct15   0:00 /usr/bin/python3 /usr/sh  
root       267  0.0  0.0   3876   896 ?        Ss   Oct15   0:00 /init  
root       268  0.0  0.0   3892  1024 ?        S   Oct15   0:03 /init  
opank     269  0.0  0.1  6072  5248 pts/0    Ss   Oct15   0:01 -bash  
root       270  0.0  0.1  6820  4480 pts/1    Ss   Oct15   0:00 /bin/login -f  
opank     302  0.0  0.2 20388 11392 ?       Ss   Oct15   0:00 /usr/lib/systemd/systemd  
opank     303  0.0  0.0  21448  3520 ?        S   Oct15   0:00 (sd-pam)  
opank     318  0.0  0.1  6056  5248 pts/1    S+   Oct15   0:00 -bash  
polkitd    855  0.0  0.1 388160  7680 ?       Ssl  Oct15   0:00 /usr/lib/polkit-1/polkit  
root      4011  0.0  0.5 370052 19968 ?       Ssl  00:56   0:00 /usr/libexec/packagekitd  
opank     4257 100.0  0.1  8200  4224 pts/0    R+   00:59   0:00 ps aux  
opank@MSI:~$ top  
top - 00:59:59 up 2:25, 1 user, load average: 0.06, 0.08, 0.02  
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie  
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st  
MiB Mem : 3766.2 total, 3858.3 free,  444.0 used,  348.5 buff/cache  
MiB Swap: 1024.0 total, 1024.0 free,  0.0 used,  3322.2 avail Mem  


| PID  | USER     | PR | NI | VIRT   | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND         |
|------|----------|----|----|--------|-------|-------|---|------|------|---------|-----------------|
| 4011 | root     | 20 | 0  | 378052 | 19968 | 17152 | S | 0.3  | 0.5  | 0:00.00 | packagekitd     |
| 1    | root     | 20 | 0  | 21956  | 13856 | 9472  | S | 0.0  | 0.3  | 0:06.36 | systemd         |
| 2    | root     | 20 | 0  | 3872   | 1792  | 1920  | S | 0.0  | 0.0  | 0:00.00 | init-systemd(Ub |
| 7    | root     | 20 | 0  | 3888   | 1920  | 1920  | S | 0.0  | 0.0  | 0:00.00 | init            |
| 50   | root     | 19 | -1 | 66832  | 18972 | 18076 | S | 0.0  | 0.5  | 0:03.11 | systemd-journal |
| 103  | root     | 20 | 0  | 25264  | 6400  | 4992  | S | 0.0  | 0.2  | 0:00.50 | systemd-udev    |
| 189  | systemd+ | 20 | 0  | 21452  | 12880 | 10024 | S | 0.0  | 0.3  | 0:00.36 | systemd-resolve |

  
opank@MSI:~$ which bash  
/usr/bin/bash  
opank@MSI:~$ which gcc  
/usr/bin/gcc  
opank@MSI:~$ ls -lh /boot/vmlinuz=$(uname -r)  
ls: cannot access '/boot/vmlinuz-6.8.87.2-microsoft-standard-WSL2': No such file or directory  
opank@MSI:~$ ls  
bash_logout      file_syscalls.c  math_ops.h       program  
hello_module.c   math_ops.i       math_ops.o       program_dynamic  
.cache           info_syscalls.c  math_ops.o       program_shared  
.config          info_syscalls.c  math_ops.s       program_static  
lessshst        libmathops.so    my_daemon        simple_shell  
local           main.c           my_daemon.c      simple_shell.c  
.motd_shown      main.i           os_services      sudo  
.profile         main.o           os_services.c    syscall_test.txt  
sudo_admin_successful  main.s          os_specific.c    test_file.txt  
Makefile         math_ops.c       process_syscalls process_syscalls.c  
file_syscalls    math_ops.c       process_syscalls.c  
opank@MSI:~$ ls  
Makefile      main.c      math_ops.i  os_specific.c  simple_shell  
file_syscalls main.i      math_ops.o  process_syscalls  simple_shell.c  
file_syscalls.c main.o     math_ops.s  process_syscalls.c  sudo  
hello_module.c main.s     my_daemon   program         syscall_test.txt  
info_syscalls  math_ops.c my_daemon.c program_dynamic  test_file.txt  
info_syscalls.c math_ops.c os_services program_shared  
libmathops.so  math_ops.h os_services.c program_static  
opank@MSI:~$ ls -lh /boot/vmlinuz=$(uname -r)  
ls: cannot access '/boot/vmlinuz-6.8.87.2-microsoft-standard-WSL2': No such file or directory  
opank@MSI:~$ cat /boot/config=$(uname -r) | grep "y" | wc -l  
cat: /boot/config-6.8.87.2-microsoft-standard-WSL2: No such file or directory  
0  
opank@MSI:~$ nano procfs_read.c  
opank@MSI:~$ gcc -o procfs_read procfs_read.c  
opank@MSI:~$ ./procfs_read  
=== CPU Information ===  
processor      : 0  
vendor_id     : GenuineIntel  
cpu family    : 6  
model         : 154  
model name    : 12th Gen Intel(R) Core(TM) i7-12650H  
stepping      : 3  
microcode    : 0xffffffff  
cpu MHz       : 2687.999  
cache size    : 24576 KB
```

9. Building and Booting an Operating System

Praktik 9.1: Analisis Boot Process

Praktik 9.2: Kernel Parameters

Praktik 9.3: Simple Bootloader (Konsep)

```
opank@MSI: ~$ systemctl --dry-run --full --show-transaction console-setup.service
5ms console-setup.service
5ms setvtrgb.service
3ms systemd-update-utmp-runlevel.service

opank@MSI:~$ systemctl analyze critical-chain
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "+" character.

graphical.target @639ms
├─multi-user.target @638ms
├─snapd.seeded.service @473ms +184ms
├─basic.target @465ms
├─sockets.target @464ms
├─snapd.socket @415ms +48ms
├─sysinit.target @411ms
├─systemd-resolved.service @340ms +78ms
├─systemd-tmpfiles-setup.service @327ms +11ms
├─local-fs.target @325ms
├─local-fs-pre.target @325ms
├─systemd-tmpfiles-setup-dev.service @320ms +48ms
├─systemd-tmpfiles-setup-dev-early.service @274ms +8ms
├─mod-static-nodes.service @238ms +15ms
├─systemd-journald.socket @205ms
├─system.slice @200ms
└─.slice @200ms

opank@MSI:~$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory
opank@MSI:~$ lsinitramfs /boot/initrd.img-${uname -r} | head -20
Command 'lsinitramfs' not found, but can be installed with:
sudo apt install initramfs-tools-core
opank@MSI:~$ sudo apt install initramfs-tools-core
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  busybox-initramfs cpio dracut-install initramfs-tools-bin klibc-utils libklibc zstd
Suggested packages:
  libarchive-dev
The following NEW packages will be installed:
  busybox-initramfs cpio dracut-install initramfs-tools-bin initramfs-tools-core
  klibc-utils libklibc zstd
0 upgraded, 8 newly installed, 0 to remove and 189 not upgraded.
Need to get 1167 kB of archives.
After this operation, 3493 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Praktik 10.5: Memory Leak Detection dengan Valgrind

[illegible]

```

00000000-~$ (gdb) break main
~bash: syntax error near unexpected token `break'

00000000-~$ dmesg
6 UTC 2025
00000000] Command line: initrd=/initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty_legacy_count=0 WSL_ENABLE_CRASH_DU
MP=1

00000000] KERNEL supported cpus:
00000000] Intel GenuineIntel
00000000] AMD AuthenticAMD
00000000] BIOS-provided physical RAM map:
00000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffffffffff] usable
00000000] BIOS-e820: [mem 0x000000000009ffffffffff-0x00000000000e00000000] reserved
00000000] BIOS-e820: [mem 0x0000000000100000-0x000000000001ffffffff] ACPI data
00000000] BIOS-e820: [mem 0x0000000000200000-0x00000000000f00000000] usable
00000000] NX (Execute Disable) protection: active
00000000] APIC: Static calls initialized
00000000] DMI not present or invalid.
00000000] Hypervisor detected: Microsoft Hyper-V
00000000] Hyper-V: privilege flag low GxHef, high 0xb3b830, hints 0x9a4e24, misc 0x0ebd7b2
00000000] Hyper-V: Nested features: 0x3e0101
00000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
00000000] Hyper-V: Using hypercall for remote I/O flush
00000000] clocksource: hyperv_clocksource.timer mask: 0xfffffffffffffff max_cycles: 0x24de6a1710, max_idle_ns: 440795202120 ns
00000000] clocksource: hyperv_clocksource.nsr mask: 0xfffffffffffffff max_cycles: 0x24de6a1710, max_idle_ns: 440795202120 ns
00000000] tsc: Detected 2687.999 MHz processor
00000000] e820 update [mem 0x00000000-0x00000000ffff] usable ==> reserved
00000000] e820 remove [mem 0x000a0000-0x00000000ffff] usable
00000000] last_pfn = 0xf6aa0 max_arch_pfn = 0x400000000
00000000] MTRR map: 5 entries (4 fixed + 1 variable; max 128), built from 8 variable MTRRs
00000000] x86/PAT Configuration [0-7]: MO UC- WC WP UC- MT
00000000] Using PG pages for direct mapping
00000000] RAMDISK: [mem 0x04b0b000-0x04e7ffff]
00000000] ACPI: Early table checksum verification disabled
00000000] ACPI: RSDP 0x0000000000000000 00002A (v02 VIRTUAL)
00000000] ACPI: XSDT 0x0000000000100000 000004 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
00000000] ACPI: FACP 0x0000000000101000 000114 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
00000000] ACPI: DSMT 0x0000000000101154 00001c (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
00000000] ACPI: FACS 0x0000000000101114 000000
00000000] ACPI: OEMO 0x0000000000101154 000004 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
00000000] ACPI: SRAT 0x00000000001f12d4 000130 (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
00000000] ACPI: APIC 0x00000000001f1600 0000C5 (v04 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
00000000] ACPI: Reserving FACP table memory at [mem 0x101000-0x101113]
00000000] ACPI: Reserving SDST table memory at [mem 0x1011b8-0x101fd3]
00000000] ACPI: Reserving FACS table memory at [mem 0x101114-0x101153]

```